

Obs sumas parciales serie fourier

convolucion nucleo dirichlet

Observación 1. Sea $f \in L^1(\mathbb{T})$, entonces las sumas parciales simétricas de su serie de Fourier vienen dadas por $\forall N \in \mathbb{Z}$:

$$\begin{aligned} S_N(f)(x) &= \sum_{n=-N}^N \widehat{f}(n) e^{2\pi i n x} = \sum_{n=-N}^N \left(\int_0^1 f(t) e^{-2\pi i n t} dt \right) e^{2\pi i n x} \\ &= \int_0^1 f(t) \left(\sum_{n=-N}^N e^{2\pi i n (x-t)} \right) dt = \int_0^1 f(t) D_N(x-t) dt = (f * D_N)(x). \end{aligned}$$

Referenciado en

- `prop-criterio-dirichlet`
- `obs-sumacion-cesaro-convolucion-nucleo-fejer`

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